

02YMECH - Solution 11

Assigned for the week of Dec 1, 2025

Solutions

1. Rod of length l and mass m , with uniform linear density

$$\lambda = \frac{m}{l}.$$

Place the rod along the x -axis from $-l/2$ to $+l/2$. The observation point lies on the axis at distance z from the center.

An element dx contributes

$$d\Phi = -\frac{G\lambda dx}{\sqrt{x^2 + z^2}}.$$

Thus,

$$\Phi(z) = -G\lambda \int_{-l/2}^{l/2} \frac{dx}{\sqrt{x^2 + z^2}}.$$

Using

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln\left(x + \sqrt{x^2 + a^2}\right),$$

we obtain

$$\Phi(z) = -\frac{Gm}{l} \ln\left(\frac{\frac{l}{2} + \sqrt{z^2 + \left(\frac{l}{2}\right)^2}}{-\frac{l}{2} + \sqrt{z^2 + \left(\frac{l}{2}\right)^2}}\right).$$

2. For a ring of radius a and mass m , a point on the axis at distance z from the center is at the same distance

$$r = \sqrt{a^2 + z^2}$$

from every mass element. Hence,

$$\Phi(z) = -G \int \frac{dm}{r} = -\frac{Gm}{\sqrt{a^2 + z^2}}.$$

3. The total mass is

$$M = \int_0^R 4\pi r^2 \rho(r) dr = \frac{4\pi\rho_0}{R} \int_0^R r^3 dr = \pi\rho_0 R^3,$$

so that

$$\rho_0 = \frac{M}{\pi R^3}.$$

The enclosed mass for $r < R$ is

$$M(r) = \frac{4\pi\rho_0}{R} \int_0^r r'^3 dr' = M \frac{r^4}{R^4}.$$

Therefore, the gravitational field is

$$\mathbf{g}(r) = \begin{cases} -GM \frac{r^2}{R^4} \hat{\mathbf{r}}, & r < R, \\ -\frac{GM}{r^2} \hat{\mathbf{r}}, & r \geq R. \end{cases}$$

4. For a uniform-density Earth of radius R and mass M , the force on a particle at distance r from the center is

$$F = -\frac{GM(r)m}{r^2} = -\frac{GMm}{R^3} r, \quad \text{where } M(r) = M \frac{r^3}{R^3}.$$

leading to the equation of motion

$$\ddot{r} = -\frac{GM}{R^3} r.$$

This is simple harmonic motion with

$$\omega = \sqrt{\frac{GM}{R^3}}.$$

The period is

$$T = 2\pi \sqrt{\frac{R^3}{GM}} \approx 5.07 \times 10^3 \text{ s} \approx 84.5 \text{ min.}$$